INCH-POUND

MIL-S-13949/25B
25 January 1993
SUPERSEDING
MIL-P-13949/25A
11 April 1988

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL CF (NONWOVEN POLYESTER, EPOXY RESIN, FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: C for nonwoven polyester.

Resin system: F in accordance with MIL-S-13949.

Special consideration (third letter designator): N or P in accordance with MIL-S-13949.

Type of metal foil: Any as specified in accordance with MIL-S-13949.

Weight of metal foil: 1, 2, or 0 in accordance with MIL-S-13949.

Solderabilty: Base materials clad with metal foil types A, B, C, D, G, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: Not applicable.

Grade of pits or dents: In accordance with MIL-S-13949.

Qualification inspection.

Extent of qualification: In accordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm)).

After bake: Class A, not applicable; class B, .0015 inch (0.040 mm) maximum.

After stress: Class A, not applicable; class B, .0020 inch (0.050 mm) maximum.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Flammability.

Duration of burning time, each flame application.

Base material thicknesses greater than or equal to .020 inch (0.51 mm).

Average: 25 seconds maximum.

Individual: 30 seconds maximum.

Base material thicknesses less than .020 inch (0.51 mm): 15 seconds maximum.

Burn length, average (for base material thicknesses less than .020 inch (0.51 mm)): 12 inches maximum.

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm)).

Crosswise: 5500 pounds per square inch minimum.

Lengthwise: 6000 pounds per square inch minimum.

Loss tangent at 1 MHz: .030 maximum.

Permittivity at 1 MHz: 4.0 maximum.

Arc resistance: 60 seconds minimum.

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 40 kilovolts minimum.

Electric strength, average, (for base material thicknesses less than .020 inch (0.51 mm)): 600 volts/mil minimum.

Surface resistivity.

After moisture resistance conditioning: 10^5 megohm minimum.

After elevated temperature (125°C) conditioning: 10^3 megohm minimum.

Peel strength values for copper foil: See table I.

TABLE I. Peel strength values, minimum pounds per inch of width.

Test and copper foil weight	Base material thickness	
	Less than .020 inch	.020 inch and over
After thermal stress:		
1-oz/ft² (designator 1)	7.0	7.0
2-oz/ft² (designator 2)	10.0	10.0
At elevated temperature of 125°C:		
1-oz/ft² (designator 1)	0.5	0.5
2-oz/ft² (designator 2)	0.5	0.5
After exposure to processing solutions:		
1-oz/ft² (designator 1)	6.0	6.0
2-oz/ft² (designator 2)	9.0	9.0

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Volume resistivity.

After moisture resistance conditioning: 10^6 megohm-cm minimum.

After elevated temperature (125°C) conditioning: 10^3 megohm-cm minimum.

Water absorption (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 1.0 percent maximum.

Characteristics data, qualification and quality conformance testing is not required.

CTE, below glass transition temperature.

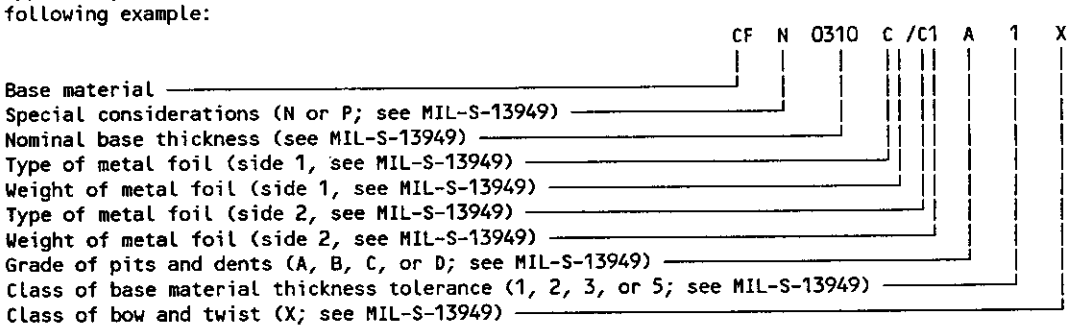
Z axis: 125 ppm/°C maximum.

X, Y axis: 40 ppm/°C maximum.

Glass transition temperature: By DSC, 25°C minimum; by TMA, 48°C minimum.

Supersession information: Not applicable.

Type designation: Consists of "CF" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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